Renzo Mancini

Curriculum Vitae,
3 years valid working visa

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Education

2021 - 2026 **PhD in Applied Mathematics**, *The University of Auckland*, New Zealand
Developed and analysed a conceptual global climate model.
- **Key skills:** Programming, visualisation, high-dimensional data, climate science, communication skills.
- **Thesis:** Bifurcation analysis of a two-delay model for the Atlantic Meridional Overturning Circulation.

2013 - 2017 **MSc in Science, Geophysics**, *University of Chile*, Chile, (graduated with Highest Honours)
Developed a 3D resistivity model of a large volcanic chain in northern Chile.
- **Key skills:** 3D spatial data analysis, surveying, inverse modelling, signal processing, multi-source data integration
- **Thesis:** Conductivity distribution beneath the San Pedro-Linzor volcanic chain, Northern Chile, using 3D magnetotelluric modelling.

2009 - 2013 **BSc in Physics**, *University of Chile*, Chile
Studied waiting times and atomic diffusion in solid-state fusion.
- **Key skills:** Computational simulation, probabilistic modelling, programming, Bash, Python.
- **Project:** Waiting times and atomic diffusion in solid-state fusion.

Experience

2021 - 2026 **Researcher in applied mathematics**, *University of Auckland*
- Developed multiple algorithms to solve and visualise complex problems in dynamical systems using Matlab, Python, Julia, VSCode and GitHub.
- Analysed global climate data and contrasted it with theoretical results, leading to two scientific papers currently under review.

2018 - 2019 **Data analyst and surveyor**, *Applied Geophysics Center (NGA)*
- Responsible for data management, data processing, and modelling.
- Built models from large geospatial datasets using Python, Excel, ArcGIS and ModEM.
- Used advanced statistical methods to solve inverse problems, resulting in the first map of the sedimentary basin in Rancagua, Chile.

2015 - 2018 **Data analyst and Team Leader**, *Andean Geothermal Center of Excellence*
- Planned geophysical research, focused on asset management, data acquisition, data analysis, and modelling.
- Participated in several MSc campaigns, helping the community and training others.

2017 - 2018 **Programmer**, *University of Chile*
- Developed an algorithm using finite element methods in Fortran and Python, results are part of a publication in a peer-reviewed journal.
- Demonstrated technical autonomy and ability to learn, by independently performing a scientific project.

2014 - 2018 **Lecturer**, *University of Chile*
- Lectured on physics and tutored master's students in geophysics, obtaining positive feedback.
- Planned courses and developed materials, demonstrating strong organisational skills.

2016 - 2025 **Conference speaker**, *Several international conferences*
- Presented my research in more than 10 international conferences, resulting in collaborative work.

2022 - 2023 **PhD Representative and seminar organiser**, *Auckland University*
- Collaborated to improve the PhD experience for students and build a stronger community.

Coding Skills

- Technologies: Python (Matplotlib, NumPy), Bash (Unix), MATLAB, Julia, Fortran, Power BI.
- Tools: GitHub, VSCode, Microsoft Office/Excel, Illustrator, LaTeX, Affinity, Topcon, ArcGIS.
- Scientific: PyDelay, DDEBiftool, ModEM, 3dGrid, Model Vision, Oasis Montaj.

Scientific Publications

Jan 2026 **R. Mancini**, B. Krauskopf, S. Russell, and H. A. Dijkstra, **Bifurcation analysis of a two-delay model for the Atlantic Meridional Overturning Circulation**, *Under review - Journal of Physics: Complexity.*

Nov 2021 L. Cabrera, S. Ruiz, P. Contreras-Reyes, A. Osses, and **R. Mancini**, **Northern Chile intermediate-depth earthquakes controlled by plate hydration**, *Geophysical Journal International 226(1): 78–90, 2021.*

May 2019 **R. Mancini**, D. Diaz, H. Brasse, B. Godoy, and M. J. Hernandez, **Conductivity distribution beneath the San Pedro-Linzor volcanic chain, North Chile, using 3-D magnetotelluric modeling**, *Journal of Geophysical Research: Solid Earth 124(5): 4386–4389, 2018.*

Conference Presentations

December 2024 **Joint Meeting of the NZMS, AustMS and AMS**, *Oral presentation: "Bifurcation analysis of a two-delay model for the Atlantic Meridional Overturning Circulation"*, Auckland, New Zealand.

August 2024 **XLIV Dynamics Days Europe**, *Oral presentation: "Bifurcation analysis of a two-delay model for the Atlantic Meridional Overturning Circulation"*, Bremen, Germany.

July 2024 **11th European nonlinear dynamics conference (ENOC)**, *Oral presentation: "Bifurcation analysis of a conceptual model with two time delays for the Atlantic Meridional Overturning Circulation"*, Delft, Netherlands.

May 2024 **Workshop on "Frontiers in Applied Dynamical Systems"**, *Poster presentation: "Bifurcation analysis of a two-delay model for the Atlantic Meridional Overturning Circulation"*, Cork, Ireland.

February 2024 **Australia and New Zealand Industrial and Applied Mathematics (ANZIAM)**, *Oral presentation: "Temporal feedback mechanism in the Atlantic Meridional Overturning Circulation, A delay-differential equation approach"*, Adelaide, Australia.

December 2023 **New Zealand mathematical colloquium (NZMS)**, *Oral presentation: "Analysis of a conceptual model for the Atlantic Meridional Overturning Circulation with two time delays"*, Wellington, New Zealand.

May 2023 **SIAM Conference on Applications of Dynamical Systems (DS23)**, *Oral presentation: "Analysis of a conceptual model for the Atlantic Meridional Overturning Circulation with two time delays"*, Portland, USA.

February 2023 **Australia and New Zealand Industrial and Applied Mathematics (ANZIAM)**, *Oral presentation: "Analysis of a conceptual AMOC model with two time delays"*, Queensland, Australia.

November 2022 **Dynamical systems in New Zealand**, *Poster presentation: "Analysis of a conceptual model for the Atlantic Meridional Overturning Circulation with two time delays"*, Castaways, New Zealand.

September 2022 **Time delay systems, International Federation of Automatic Control (IFAC)**, *Oral presentation: "Analysis of a conceptual model for the Atlantic Meridional Overturning Circulation with two time delays"*, Montreal, Canada.

January 2019 **Inverse Problems Methods, Applications and Synergies (2019)**, *Poster presentation*, 2-D thermal model of the Chilean subduction zone using the finite element method
Santiago, Chile

November 2019 **XV Chilean Geological Congress (XV CGCH2018)**, *Oral presentation*, Distribution of conductive elements beneath the San Pedro–Linzor volcanic chain, northern Chile, using 3-D magnetotelluric modeling
Concepción, Chile

August 2018 **XXIV Electromagnetic Induction Workshop (EMIW2018)**, *Poster presentation in collaboration with Dr. Daniel Díaz*, Conductivity distribution of magmatic and hydrothermal structures related to the San Pedro–Linzor volcanic chain, northern Chile, using 3-D magnetotelluric modeling
Helsingør, Denmark

December 2017 **AGU Fall Meeting**, *Poster presentation in collaboration with Dr. Daniel Díaz*, 3-D modeling of magnetotelluric data in the Paniri–Toconce volcanic chain, Central Andes
New Orleans, United States

March 2017 **Third Colloquium on Earthquakes and Volcanoes (3gsev)**, *Poster presentation in collaboration with Dr. Daniel Díaz*, Three-dimensional conductivity model of the Paniri–Toconce volcanic system, Chile
Concepción, Chile

November 2016 **IX Cities on Volcanoes (COV)**, *Poster presentation in collaboration with Dr. Daniel Díaz*, Resistivity structure of the Paniri–Toconce volcanic system, Chile
Puerto Varas, Chile

July 2016 **XXIII Electromagnetic Induction Workshop (EMIW2016)**, *Poster presentation in collaboration with Dr. Daniel Díaz*, Three-dimensional modeling of the Paniri–Toconce volcanic system using magnetotellurics
Chiang Mai, Thailand

May 2016 **Second Colloquium on Earthquakes and Volcanoes (2gsev)**, *Oral presentation in collaboration with Dr. Daniel Díaz*, Looking for magmatic reservoirs beneath the Paniri–Toconce volcanic complex, Department of Geophysics, University of Chile
Santiago, Chile

November 2012 **XVIII Chilean Symposium on Physics**, *Poster presentation of a project under the supervision of Ph.D. Sergio Davis*, Waiting times and atomic diffusion in solid fusion, Nanomaterials Group (www.gnm.cl)
La Serena, Chile

Research

2021–2026 **Bifurcation analysis of a two-delay model for the AMOC**, *Developed and analyzed a conceptual global climate model via bifurcation analysis using MATLAB and Python.*, Department of mathematics, Auckland University, New Zealand

2018–2019 **Field expert and data analyst in the "Colegio-Araucanía" project**, *ERT field team technician*, Andean Geothermal Center of Excellence (CEGA), University of Chile, Santiago

January 2018 – November 2018 **Study of the Rancagua Basin through Gravimetric and Transient Electromagnetic Analysis**, Data processing, analysis, and acquisition. Applied Geophysics Group, Department of Geophysics, Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

March 2018 – December 2018 **Thermal modeling of the northern Chile subduction zone using numerical methods**, Thermal model programming with finite elements. FONDECYT project "Seismic structure of the Nazca and Iquique Ridges and implications for hotspot magmatism and ridge-trench collision along the South American subduction zone", project FONDECYT 1170009, Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

March 2016 – 2017 **Three-dimensional modeling of the Paniri-Toconce volcanic system using magnetotellurics, FONDAP Project 15090013**, Analysis, data acquisition, and interpretation of Magnetotelluric (MT) stations, in collaboration with the Andean Geothermal Center of Excellence (CEGA), Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

March 2015 – July 2015 **Study of the Santiago Basin through Gravimetric Analysis**, Data acquisition and analysis under the supervision of Dr. Andrei Maksymowicz and Dr. Emilio Vera, supporting the Applied Geophysics Group, Department of Geophysics, Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

2012 **Waiting times and atomic diffusion in solid fusion**, Research collaborator with Ph.D. Sergio Davis, Nanomaterials Group (www.gnm.cl), Department of Physics, Faculty of Sciences, University of Chile, Santiago

March 2011 – July 2011 **Study of atomic diffusion in solid fusion using the z-method**, Research unit under the supervision of Ph.D. Gonzalo Gutiérrez and Ph.D. Sergio Davis, supported by the Nanomaterials Group (www.gnm.cl), Department of Physics, Faculty of Sciences, University of Chile, Santiago

Research visits

March 2026 **Auckland University, New Zealand**, I worked on finishing a paper on a conceptual study of the Atlantic Meridional Overturning Circulation and began a second paper describing a new type of dynamics in a double-delay system, with the support of Bernd Krauskopf.

March 2017 **Freie Universität Berlin, Germany**, I reanalyzed data from low-frequency band MT stations, which led to a subsequent publication. Additionally, I participated in several talks.

Data Acquisition Campaigns

July 2019 **CONICYT PIA/Research Ring in Science and Technology ACT172002**, Magnetism data acquisition campaigns in the Antofagasta Region, Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

March 2019 **Applied Geophysics Group**, *Acquisition of Transient Electromagnetic (TEM) and gravity data in the Libertador General Bernardo O'Higgins Region*, NGA

January 2019 **FONDAP Project 15090013**, ERT data acquisition in support of the Andean Geothermal Center of Excellence (CEGA), Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

March 2018 **FONDAP Project 15090013**, Transient Electromagnetic (TEM) data acquisition campaign carried out in the El Tatio geothermal field, Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

June 2017 **Rajo Inca CODELCO Project**, *Field leader in Transient Electromagnetic (TEM) data acquisition*, Idiem

March 2017 **International Project Laguna del Maule (NSF)**, Magnetotelluric (MT) data acquisition campaign in support of Darcy Cordell, PhD student at the University of Alberta, in the eastern sector of Laguna del Maule

March 2016 and January 2017 **FONDAP Project 15090013**, Two Magnetotelluric (MT) data acquisition campaigns in the vicinity of the Paniri-Toconce volcanic chain, Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

October 2016 **Master's Thesis Support**, Fourth Magnetotelluric (MT) data acquisition campaign carried out in support of the thesis of Diego Naranjo, in the vicinity of San José de Maipo volcano, Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

October 2015 **Master's Thesis Support**, First Magnetotelluric (MT) and Transient Electromagnetic (TEM) data acquisition campaign carried out in support of the thesis of Diego Naranjo, in the vicinity of San José de Maipo volcano, Faculty of Physical and Mathematical Sciences (FCFM), University of Chile, Santiago

Teaching Experience

At University of Chile **Lecturer**
Physics I and II– Mechanics, thermodynamics, and electromagnetism for architecture students, 2014–2018: Course lecturer to ∼60 undergraduate students. Set office hours, in charge of designing the course program, including objective, class content, test, and exam questions.

Lecturer
Physics I and II– Mechanics and electromagnetism for Design students, 2014–2018: Course lecturer to ∼80 undergraduate students. Set office hours, in charge of designing the course program, including objective, class content, test, and exam questions.

University **Teaching Assistant**, *Tutor*
Physics of the Earth – 2015 and 2018: ∼15 Master students. University of Chile.
Seismology – 2015 – 2017: ∼20 Master students. University of Chile.
Geophysical Prospecting II – 2016, semester 2: ∼10 Master students. University of Chile.
Geophysical Exploration Methods – 2015 – 2016: ∼30 undergraduate students. University of Chile.
Physics – 2014: ∼30 undergraduate students. Military Polytechnic.
Mathematics I – 2014: ∼80 undergraduate students. University of Chile.
Mechanics II – 2013: ∼60 undergraduate students. University of Chile.
Mechanics I – 2013: ∼60 undergraduate students. University of Chile.
Introduction to Mechanics – 2012: ∼60 undergraduate students. University of Chile.

School **Secondary grade teacher**
Physics – 2018: Course lecturer to ∼30 school students. In charge of designing the course program, including objectives, class content, tests, and exam questions.

Administrative experience

2024 **Support to Conference Organisers**, *Joint Meeting of the NZMS, AustMS and AMS, Auckland*
I assisted in ensuring a smooth running of the conference by helping with the management of registration, guidance of participants, and ensuring the proper setup and functioning of the rooms for talks and sessions.

2022–2023 **PhD representative**, *University of Auckland*, New Zealand
My role as PhD representative includes organizing events that foster connection and community among PhD students, providing opportunities for them to meet and support one another. I also serve as a point of contact for students' needs, ensuring their concerns are heard and addressed. Additionally, I participate in meetings with faculty and administration to advocate for students and contribute to improving the overall PhD experience.

2021–2022 **Organiser: Fortnightly Postgraduate Seminar**, *University of Auckland*, New Zealand
The seminar provides a place for PhD students to present their research projects, fostering an inclusive environment that welcomes diverse subfields such as mathematics education, applied mathematics, and pure mathematics. This initiative aimed to promote community building, collaboration, and knowledge exchange in the Department of Mathematics.

Outreach activities

2018 **Expo physics**
Talk at the University of Chile: Geophysical methods, Santiago, Chile.

2017 **Expo physics**
Talk at the University of Chile: Geophysics as science, Santiago, Chile.

2016 **Expo physics**
Talk at the University of Chile: Geophysical exploration using magnetotelluric, Santiago, Chile.

Grants

2023 I was awarded the Performance-Based Research Fund (PBRF) from the Department of Mathematics of the University of Auckland to support my attendance at the SIAM DS23 conference.

2018 I was awarded a full travel and registration grant from the XXIV Electromagnetic Induction Workshop (EMIW2018).

2017 I was awarded a full travel grant from the University of Chile to undertake a four-month research visit at Freie Universität Berlin, Germany, as part of my master's research.